

CHAPTER 11

11.1) CHEMICAL POTENTIALS IN TWO-PHASE EQUILIBRIUM

The chemical potentials for atomic species A and B are given by

$$\mu_A = (\partial F / \partial N_A)_{N_B} \quad , \quad \mu_B = (\partial F / \partial N_B)_{N_A} \quad . \quad (S1)$$

Using (1) - (3), we write the free energy F in the form

$$F(x) = Nf(x) = (N_A + N_B)f(x) \quad , \quad (S2)$$

where

$$x = N_B / N = N_B / (N_A + N_B) = 1 - N_A / (N_A + N_B) \quad .$$

Note that

$$(\partial x / \partial N_A)_{N_B} = - N_B / (N_A + N_B)^2 = - x / N \quad , \quad (S3)$$

$$(\partial x / \partial N_B)_{N_A} = + N_A / (N_A + N_B)^2 = (1-x) / N \quad . \quad (S4)$$

Combining (S1) through (S4):

$$\mu_A = f(x) - Nf'(x)(\partial x / \partial N_A) = f(x) - xf'(x) \quad ,$$

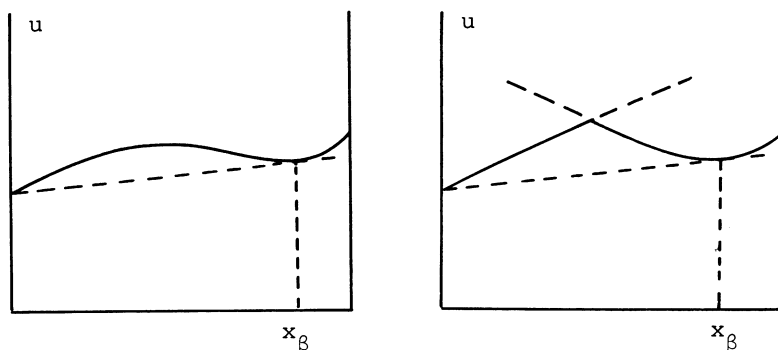
$$\mu_B = f(x) - Nf'(x)(\partial x / \partial N_B) = f(x) + (1-x)f'(x) \quad .$$

From these expressions one sees readily that μ_A and μ_B are simply the two intercepts, of the vertical lines $x = 0$ and $x = 1$, with the tangent to the continuous curve $f(x)$, taken at the composition x of interest. For a two-phase equilibrium we saw that the two compositions x_α and x_β belong to two points with a common tangent, hence μ_A and μ_B are the two intercepts with this two-point tangent.

Comment. Our derivation shows that the two intercepts of any tangent to the $f(x)$ -curve give the two chemical potentials μ_A and μ_B for the two constituents of the single phase represented by $f(x)$, for the composition x corresponding to the tangential point. Applied to a two-phase equilibrium the common-tangent rule is therefore evidently equivalent to the equilibrium requirement that for each constituent separately the chemical potential must be the same in both phases. Different constituents may, of course, have different chemical potentials.

11.2) MIXING ENERGY IN $^3\text{He} - ^4\text{He}$ AND Pb-Sn MIXTURES

The curve $u(x)$ must have a shape qualitatively similar to the two curves shown below, permitting a straight line to be drawn that intersects $u(x)$ at $x = 0$ and that is tangential to $u(x)$ at $x \cong 0.94$ ($^3\text{He}-^4\text{He}$) or at $x \cong 0.97$ (Pb-Sn).



11.3) SEGREGATION COEFFICIENT OF IMPURITIES

When $x \ll 1$ we may expand the term $(1-x)\log(1-x)$ in the mixing entropy, Eq. (11), by setting $(1-x) \cong 1$, $\log(1-x) \cong -x$. This leads to a mixing entropy per atom,

$$\sigma_M/N \cong x - x \log x = x(1 - \log x) \quad ,$$

and to a total free energy per atom

$$f(x) = f_0(x) + \tau \sigma_M / N$$

$$= f_0(0) + x[f'_0(0) + \tau(1 - \log x)] \quad .$$

The derivative of this is

$$f'(x) = f'_0(0) + \tau(1 - \log x) - \tau = f'_0(0) - \tau \log x \quad . \quad (S1)$$

An expression of this form, with different values of $f'(0)$, applies to both the liquid and the solid phase.

The liquid and the solid phases will be in equilibrium when the two free energy curves $f_L(x)$ and $f_S(x)$ have a common tangent and the two compositions are those corresponding to the tangential points:

$$f'_L(x_L) = f'_S(x_S) \quad .$$

With (S1) and with $k \equiv x_S/x_L$, this may be written

$$f'_{0L}(0) - f'_{0S}(0) = \tau \log(x_S/x_L) = \tau \log k$$

or

$$k = \exp(-w/\tau) \quad , \quad \text{where } w = f'_{0S} - f'_{0L} \quad .$$

For the numbers specified, $k \cong \exp(-11.6) \cong 9.1 \times 10^{-6}$. Segregation coefficients this small, although not common, do occur.

11.4) SOLIDIFICATION RANGE OF A BINARY ALLOY

Qualitatively, the claimed result is all but obvious: When the last infinitesimal remnant of liquid solidifies, the solid forming from this remnant must have the same composition, but the two (equal) compositions must also be in equilibrium with each other. This can take place only at the point where the liquidus and solidus curves merge, at $x = 0$, $T = T_A$.

Extension. It is instructive to extend the problem and its discussion along the following lines. Let

$$x = N_B/N = 1 - N_A/N$$

be the fraction of B-atoms in the liquid phase, with N_A , N_B and $N = N_A + N_B$ all referring to the liquid phase. Thus, if N_A and N_B change by dN_A and dN_B , we have

$$dx = - \frac{N_B dN_A}{N^2} + \frac{N_A dN_B}{N^2} = -x \frac{dN_A}{N} + (1-x) \frac{dN_B}{N} \quad . \quad (S1)$$

If the changes dN_A and dN_B are negative (solidification), the ratio dN_A/dN_B is governed by the solidus curve of the phase diagram,

$$dN_A = (1-x_S)dN \quad , \quad dN_B = x_S dN \quad , \quad (S2)$$

where x_S is the fraction of B-atoms in the solid in equilibrium with the liquid, at the temperature in question. Inserting (S2) into (S1) leads to

$$dx = (x_S - x)dN/N \quad .$$

Division by $x_S - x$ and integration:

$$\int \frac{dx}{x_S - x} = \int \frac{dN}{N} = \log(N/N_0) \quad . \quad (S3)$$

Here N_0 designates the number of atoms initially present as liquid phase, and N the number of atoms left as liquid after partial solidification.

To proceed further, we must make some assumption about the interrelation between x_S and x . The simplest possibility is $x_S = kx$, where k is a constant proportionality factor, called a segregation coefficient as in the preceding problem. In the present problem, $k > 1$. For sufficiently small values of x , a constant value of k always becomes a good approximation, because near $x = 0$ both the liquidus

and the solidus curves may always be approximated as straight lines intersecting at $x = 0$, and k is simply the solidus-to-liquidus slope ratio.

If we insert $x_s = kx$ into (S3), the integration on the left side is readily performed, and the result may be re-arranged to read

$$x/x_0 = (N/N_0)^{k-1} \quad , \quad (S4)$$

where x_0 is the initial liquid composition parameter. The composition of the concurrently forming solid is of course $x_s = kx_0(N/N_0)^{k-1}$. The temperature at which this solidification takes place is readily obtained from the slope of the liquidus curve in the phase diagram.

Note that the entire calculation depends on the assumption that the liquid phase is homogeneous (through either mixing or diffusion), while diffusion in the solid is negligible, and the liquid is in diffusive equilibrium only with that part of the solid phase that was solidified last.

It is clear from (S4) that for $k > 1$, the concentration of the B-atoms in the liquid phase decreases with decreasing amount of liquid, and if $k \gg 1$ the decrease will be very rapid. Our formalism also applies to $k < 1$, in which case (S4) predicts an enrichment of B-atoms in the liquid phase. However, in this case the assumption of a constant value of k must ultimately break down, and (S4) is valid only for the initial part of the solidification process.

The segregation processes discussed in this problem play a large role in the technology of crystal growth, such as the growth of semiconductor crystals.

11.5) ALLOYING OF GOLD INTO SILICON

(a) We introduce the following symbols

t_{Au} : thickness of deposited Au film (10^{-5} cm)
 t_{Si} : thickness of dissolved Si
 ρ_{Au} & ρ_{Si} : densities of Au and Si ($19.3, 2.33 \text{ g cm}^{-3}$)
 M_{Au} & M_{Si} : atomic weights of Au and Si ($197, 28 \text{ g mol}^{-1}$)
 x_{Au} & x_{Si} : atomic fractions of Au and Si in liquid Au-Si alloy.

Then $n_{\text{Au}} = t_{\text{Au}} \rho_{\text{Au}} / M_{\text{Au}}$ and $n_{\text{Si}} = t_{\text{Si}} \rho_{\text{Si}} / M_{\text{Si}}$ are the numbers of moles of Au and Si atoms per unit area, contained in the liquid alloy film. We must have $n_{\text{Au}} = x_{\text{Au}} n$ and $n_{\text{Si}} = x_{\text{Si}} n$, where n is the total number of moles (Au + Si) per unit area in the liquid. The condition $n_{\text{Si}} / x_{\text{Si}} = n_{\text{Au}} / x_{\text{Au}}$ ($= n$) may be written, with the above values for n_{Au} and n_{Si} :

$$t_{\text{Si}} = t_{\text{Au}} \frac{\rho_{\text{Au}} M_{\text{Si}} x_{\text{Si}}}{\rho_{\text{Si}} M_{\text{Au}} x_{\text{Au}}} .$$

At $T = 400^{\circ}\text{C}$ we have, from Fig. 11.11: $x_{\text{Si}} \cong 0.32$, $x_{\text{Au}} = 1 - x_{\text{Si}} \cong 0.68$. (Note that we must be on the Si-rich branch of the solidus curve, because the melt is in equilibrium with a large Si reservoir, not with Au). If we insert our numerical values we obtain $t_{\text{Si}} = 554 \text{ \AA}$.

(b) At $T = 800^{\circ}\text{C}$: $x_{\text{Si}} \cong 0.45$, $x_{\text{Au}} \cong 0.55$, whence $t_{\text{Si}} \cong 963 \text{ \AA}$.